

BioSpecInfo

Software Bill of Materials (SBOM)

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Application version	bsi-v191
Commit	801ca3ee95bb1a90d808999a0671b3453f5d5266
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Machine-readable format	<code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	<code>RDKit_minimal.js</code>	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	<code>RDKit_minimal.wasm</code>	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	<code>three.min.js</code>	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	<code>three_bloom.js</code>	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	<code>gltf_loader.js</code>	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	<code>3Dmol-min.js</code>	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).
SmilesDrawer _{Daniel Probst}	<code>smiles-drawer.min.js</code>	MIT	246 kB	2D structure drawing from SMILES.
sql.js (JS glue) _{sql.js / SQLite}	<code>lib/sql-wasm.js</code>	MIT	45 kB	In-browser SQLite database for the data exercises.

Component	Version/File	Licence	Size	Role
SQLite (WebAssembly) _{SQLite Consortium}	lib/sql-wasm.wasm	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	lib/dimuon.js	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
RDKit_minimal.js	58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9
RDKit_minimal.wasm	e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
glTF_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d

2. Code written for this project

File	Size	Role
index.html	4690 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	35 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-prettsch.js	37 kB	The NMR estimation tables transcribed in full from Prettsch-Bühlmann-Badertscher, 4th ed.: ¹³ C and ¹ H benzenes, ethylenes, alkanes, alkynes, ¹³ C aliphatics, steric corrections, ¹ J(C,H). Numbers only, no code.
bsi-nmr.js	56 kB	Per-atom ¹ H and ¹³ C NMR prediction: applies the bsi-prettsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.

File	Size	Role
<code>bsi-nmr2d.js</code>	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
<code>bsi-moldraw.js</code>	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
<code>bsi-spettrotlettore.js</code>	53 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
<code>bsi-mol3d.js</code>	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
<code>bsi-lingue.js</code>	159 kB	Internationalisation engine: fourteen languages, writing direction included.
<code>bsi-molingue.js</code>	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
<code>bsi-pannelli-lingua.js</code>	66 kB	Panels for languages and chemical languages.
<code>sw.js</code>	5 kB	Service Worker: offline operation and network strategy.
<code>rdkit_lab.html</code>	260 kB	Cheminformatics laboratory.
<code>astro.html</code>	2660 kB	Astrochemistry module.
<code>chimorga.html</code>	194 kB	Organic chemistry module.
<code>proxy/spectra-proxy.js</code>	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
<code>index.html</code>	4241b66609ed25cbed9b9a0305309f44c033cc677ba37fce2e8cf5b859e4221
<code>bsi-ai-hub.js</code>	985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476
<code>bsi-spettri.js</code>	be757f15a70f884e3374d401596e2dfd18bcc852cafb4584920e2b2d24eb3c9
<code>bsi-cheminfo.js</code>	59a3111b3b9436add5376fde1be6f90ded3dc08c095cebd04da50c7f8268bbc
<code>bsi-prettsch.js</code>	27f727240363579fba3dba50ab3b567fbcd026a8a9a06c0e43968b195c479db3
<code>bsi-nmr.js</code>	c090d614ed5d7ed1fb958f6efec1fc4021cdc7b0f4fd6e1ca89bba167528d259
<code>bsi-geom3d.js</code>	a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054
<code>bsi-nmr2d.js</code>	a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78
<code>bsi-moldraw.js</code>	45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e
<code>bsi-spettrotlettore.js</code>	dbdc79aa8e6fc14d26ae9633f15598eb7121efd6e1cf0118e5b9ff99681c567c
<code>bsi-mol3d.js</code>	93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893
<code>bsi-lingue.js</code>	588beac68e5ea9fcfdc430a3b56290eb61df509a41c507aa24ba8c146a809431

File	SHA-256
bsi-molingue.js	e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb
bsi-pannelli-lingua.js	4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93
sw.js	5d4d70433d8ff0e40bc3c7272edec179876e2ded86d7c5569202fbd17a400f8e
rdkit_lab.html	066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad
astro.html	3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d
chimorga.html	047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d
proxy/spectra-proxy.js	808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
playwright-core	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

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